

Standards for Official statistics on Climate-Health interactions

Indicator 1.1 Mortality attributed to non-optimal temperature: Methodology

Indicator 1.2 Hospitalisations attributed to non-optimal temperature: Methodology

Note, it has been decided that indicator 1.2 does not require a separate Methodology document due to very similar methods applied. This document will need to be extended to incorporate any specific considerations for indicator 1.2.

Reproducible code

The code library for the calculation of this indicator can be found on GitLab. It can be adapted to users' own needs and data requirements.

1. Overview

1.1. Purpose

The indicator assesses the short-term impact of non-optimal temperature on mortality by providing estimates of local excess risk of death attributable to non-optimal outdoor temperature. This can give insights into changes in mortality over time and differences between localities. Disaggregation of the indicator can also help evidence differences across regions and vulnerable groups with highest risk of mortality from non-optimal temperature and inform local adaptation needs.

There are three main outcomes calculated for this indicator:

- Relative risk for the temperature-mortality association for each locality
- Number of deaths attributable to non-optimal outdoor heat and cold for each locality
- Deaths per 100,000 population attributable to non-optimal outdoor heat and cold for each locality

1.2. Concepts & definitions

Attributable number (AN):

The estimated number of deaths that are attributable to non-optimal temperatures, that is, that would not have occurred if there had been no exposure to hot or cold days.

Attributable rate (AR):

The estimated number of deaths per 100,000 population that are attributable to non-optimal temperatures, that is, that would not have occurred if there had been no exposure to hot or cold days.

Relative risk (RR):

The likelihood of an individual dying during, or shortly after, exposure to a certain daily temperature. When a temperature has a RR of 1, this means there is neither an increase nor a decrease in the likelihood of the individual dying during, or shortly after, exposure to that temperature. A RR greater

than 1 indicates an increased likelihood of death compared with the reference temperature, the MMT.

Minimum mortality temperature (MMT):

The temperature at which the relative mortality risk in a particular country or locality is lowest. For each year the indicator is calculated, a fixed number of previous years (5-15 years) is used to determine the percentiles

Optimal temperature range (OTR):

The temperature range at which mortality risk is lowest. It is defined as the range around the MMT where the RR is between 1 and 1.1. The mortality following days with temperatures in the OTR is not considered attributable to non-optimal temperature in this method. Although the risk of mortality is low on days with temperatures in the OTR, there may still be a substantial proportion of associated temperature-related deaths. A small increase in RR can contribute a substantial number of yearly deaths if many days have a temperature within that temperature range.

Hot days:

Individual days where the temperature is considered non-optimally hot. There is no global standardised definition for hot days which this indicator draws on. The daily maximum temperature is used for their determination, identifying days where the maximum temperature is above a threshold for extreme heat. Moderate-intensity hot days are defined as days where the maximum temperature is between the maximum of the OTR and the 97.5th percentile of the timeseries' daily maximum temperature distribution. High-intensity hot days are days where the temperature is above the 97.5th percentile. For each year the indicator is calculated, a fixed number of previous years (5-15 years) is used to determine the percentiles. Depending on the temperature distribution, the OTR may contain the 97.5th percentile. In this case, the upper limit of the OTR is used as the threshold between optimal temperatures and hot days.

Hot nights:

Individual nights where the temperature is considered non-optimally hot. There is no global standardised definition for hot nights which this indicator draws on. The daily minimum temperature is used for their determination, identifying nights where the minimum temperature is above a threshold for extreme heat. Moderate-intensity hot nights are defined as nights where the minimum temperature is between the maximum of the OTR and the 97.5th percentile of the timeseries' daily minimum temperature distribution. High-intensity hot nights are nights where the temperature is above the 97.5th percentile. For each year the indicator is calculated, a fixed number of previous years (5-15 years) is used to determine the percentiles. Depending on the temperature distribution, the OTR may contain the 97.5th percentile. In this case, the upper limit of the OTR is used as the threshold between optimal temperatures and hot nights.

Cold days:

Individual days where the temperature is considered non-optimally cold. There is no global standardised definition for cold days which this indicator draws on. The daily maximum temperature is used for their determination, identifying days where the maximum temperature is below a threshold for extreme cold. Moderate-intensity cold days are defined as days where the maximum temperature is between the minimum of the OTR and the 2.5th percentile of the timeseries' daily maximum temperature distribution. High-intensity cold days are days where the temperature is below the 2.5th percentile. For each year the indicator is calculated, a fixed number of previous

years (5-15 years) is used to determine the percentiles. Depending on the temperature distribution, the OTR may contain the 2.5th percentile. In this case, the lower limit of the OTR is used as the threshold between optimal temperatures and cold days.

Cold nights:

Individual nights where the temperature is considered non-optimally cold. There is no global standardised definition for cold nights which this indicator draws on. The daily minimum temperature is used for their determination, identifying nights where the minimum temperature is below a threshold for extreme cold. Moderate-intensity cold nights are defined as nights where the minimum temperature is between the minimum of the OTR and the 2.5th percentile of the timeseries' daily minimum temperature distribution. High-intensity cold nights are nights where the temperature is below the 2.5th percentile. For each year the indicator is calculated, a fixed number of previous years (5-15 years) is used to determine the percentiles. Depending on the temperature distribution, the OTR may contain the 2.5th percentile. In this case, the lower limit of the OTR is used as the threshold between optimal temperatures and cold nights.

Heatwaves:

The [UNDRR-ISC Hazard Definition and Classification Review](#) defines a heatwave as “a marked unusual period of hot weather over a region persisting for at least two consecutive days during the hot period of the year based on local climatological conditions, with thermal conditions recorded above given thresholds”. This definition is used to identify heatwaves as at least two consecutive hot days, where hot days are defined as above. This framework does not restrict heatwaves to the hot period of the year to ensure a full representation of the temperature impact on health throughout the year.

Distributed lag non-linear model (DLNM):

A model often used to describe an exposure-time-response relationship where the exposure effect is non-linear and can have delayed effects.

Spline function:

A function defined piecewise by polynomials, where each polynomial is joined to another at a fixed point, a knot.

Quasi-Poisson regression:

A quasi-Poisson distribution is used for count data, where the variance of the data is assumed to be a linear function of the mean. This distribution tends to be used when a count variable is over-dispersed.

2. Data and variables

2.1. Data template

The [data template](#) provides a list of the variables needed for the indicator, identifying those which are mandatory for the method to work, and those which are optional. Their format is also provided to align with the data requirements for the indicator code provided in GitLab. The codes and variable groupings are, where appropriate, aligned with the [Inter-agency and Expert Group on Sustainable Development Goal Indicators' Statistical Data and Metadata Exchange](#)..

This methodology uses daily data and weekly data, with the method adapted accordingly, if daily ones are not available. The recommended timeseries length is a minimum of 5 years.

Data are needed for daily measures of possible confounders between the temperature and mortality relationship, namely, for wind speed, air pressure, humidity, air pollution, and day of the week. These variables are adjusted for in the analysis. Day of the week is adjusted for because in many countries, deaths are only available by date of registration rather than date of death and fewer deaths are likely to be registered on weekends even though they might occur on those days. The other potential confounders have a bidirectional association with temperature – e.g., extreme heat can lead to increased humidity and vice versa.

The data should also contain variables for the number of deaths and the population of every disaggregation. For example, if we wish to disaggregate by sex, we need variables containing the number of male deaths, the size of the male population, the number of female deaths and the size of the female population.

2.2. Data sources

Official data sources are used where possible:

- Daily mortality and demographic data: from national administrative records
- Daily mean temperature and other climate data: from governmental meteorological offices

The following high-quality global datasets are given as proxy data sources in case of missing administrative data:

- Health and Demographic surveillance systems (HDSSs): for mortality data.
 - These are population-based health and vital event registration systems that monitor demographic and health events in a geographically defined population at regular intervals.
- [ERA5](#): for mean temperature and other climate data.
 - It is a source of global ground level temperature data which provides hourly estimates of many atmospheric, land and oceanic climate variables. The data are of high accuracy, even though they are partially modelled. ERA5 provide reanalysis data which combines observations with modelled data to have spatial coverage in the grid. This is especially useful where ground level data is patchy.
- [Air Quality Modelling Products | 4 Earth Intelligence](#): for air pollution data.
 - 4 Earth Intelligence's (4EI) Air Quality product provides satellite-derived modelled air quality data at 20m resolution globally.

For each data source used, it is important to note how the data are gathered, and thus, their limitations and accuracy.

2.3. Data preparation

All data, the mortality, demographic, and climate data, are joined to create a daily timeseries dataset by locality. Each variable is checked for duplicates, errors, and missing entries and is formatted as described in the data template.

To deal with missing climate data, the data can be imputed. Dropping observations is not appropriate as it would remove a point in the timeseries. An appropriate imputation method is interpolation for each locality as it takes autocorrelation of timeseries data into account. If an

optional variable has very few or no observations for a locality, this variable can be dropped from the equation or imputed with the daily mean of the non-missing regional values.

2.4. Data limitations

There is often a delay between death occurrences and death registrations. Providing timely statistics can, therefore, compromise the completeness of mortality data. A balance needs to be found between timeliness and completeness.

Local temperature data may be estimates taken from nearby areas, thus reducing the accuracy of such data. If the temperature data contain estimates, this should be recognised in the reporting of the output.

3. Analytical methodology

3.1. Introduction to methodology

The statistical approach for this indicator is based on, and adapted from a [method developed by Gasparrini et al.](#) which uses DLNMs to estimate the excess risk of death attributable to non-optimal outdoor temperature. This approach was adopted for this indicator because DLNMs account for the non-linear and lagged relationship between temperature and mortality.

This method uses daily, local data. However, where a daily breakdown is unavailable, weekly aggregations of temperature and mortality data may be used. Weekly data have been shown to approximate the outcomes from daily timeseries in Europe (Ballester, et al., 2023). Additionally, if local data is of poor quality or unavailable, sub-national data may be used instead.

Local breakdowns are recommended as temperature and health outcomes may vary substantially between localities and granular geographic breakdowns in the output provide a more detailed understanding of local impact. Thus, adaptation needs can be more precisely targeted. However, when calculating the indicator with the various disaggregations described in section 3.1.1, it is recommended this be calculated using regional-level data as local numbers are likely too small to give sensible results.

The indicator covers mortality from all causes to ensure that it is applicable when cause-specific mortality data is not yet available. However, where such data exist, the indicator is broken down by certain causes of death so that the physiological impact of non-optimal heat and cold can be identified.

3.1.1. Disaggregations

Table 1 describes the various disaggregations to compare the indicator's output by. These are factors which may modify the effect of extreme temperature on health outcomes. They are disaggregated by and not simply adjusted for to understand differences in trends in health outcomes between these groups and identify vulnerable groups in each locality.

The [Framework Introduction](#) document describes the recommended classifications to use for each of these disaggregations. However, the cause of death disaggregation is particular to this indicator.

The RR, AN and AR are calculated for each grouping in these disaggregations to identify changes in risk and health outcome between groups. This can be done by changing the dependent and population variables for the analysis to be the deaths and the population size for the relevant group.

Table 1: indicator disaggregations

Disaggregation	Application
Age	Using the classification described in the Framework Introduction for age at time of death
sex	Using the classification described in the Framework Introduction for sex of the deceased
Urbanisation	Using the classification described in the Framework Introduction for the degree of urbanisation of the reference area of the deceased's place of death or residence
Socio-economic group	Using the classification described in the Framework Introduction for the socio-economic group of the deceased or for the deceased's place of residence
Ethnic group	Using the classification described in the Framework Introduction for the deceased's ethnicity
Occupation	Using the classification described in the Framework Introduction for the deceased occupation
Cause of death	Disaggregate by: • heatstroke and other heat illnesses (for non-optimal heat); • hypothermia (for non-optimal cold); • cardiovascular diseases; • respiratory disease; • kidney diseases; and • diabetes These are the causes of death which often account for the highest number of deaths following non-optimal heat and cold.

3.2. Descriptive statistics

Certain basic descriptive statistics for the analysis' variables provide insights into the quality and validity of the data.

- The **minimum and maximum** of numerical variables. To indicate the credibility of the variable's range.
- The **proportion of missing entries**. To indicate whether imputation is needed to complete the timeseries or whether too many data are missing for a variable to be informative.
- **Correlation coefficients**
 - **between daily temperature and deaths**. To provide insight into the strength of the relationship between mortality and temperature.
 - **between potential confounders and between effect modifiers**. To provide insight into how the indicator's outputs can be interpreted.
- **Timeseries plots with moving averages** for deaths and the independent variables. To show high-level patterns and long-term trends.
- **Scatterplots** of deaths against the independent variables. To show the relationship between deaths and the predictors.
- **Histogram of temperature** data. To identify anomalies in the temperature data distribution – it should resemble a normal distribution.

3.3. Approach to analysis

3.3.1. Steps in the analysis

For each locality, the RR of the temperature-mortality association is determined, and the AN and the AR of deaths from non-optimal temperatures in a year are calculated. The analysis is repeated with a sliding 5- to 15-year timeseries for each year for which the RR, AN and AR are produced. If the timeseries is long enough, a 15-year timeseries is recommended. However, 5 years is the minimum number of years on which the calculations should be based. Additionally, to smooth out the estimates and be able to identify trends in time more easily, the AN and AR for a year are calculated by averaging the ANs and ARs from the 5 latest years in the timeseries. For example, if the AN and AR are calculated for 2019, years 2005-2019 (or minimum 2015-2019 if enough data is not available) are used for the input data. The output for 2019 is then calculated by averaging the ANs/ARs for years 2015-2019.

The AN and AR are calculated only for non-optimal temperatures, and not for the OTR, because causality between temperature and mortality is more easily assumed for temperature extremes than for milder temperatures. This indicator is calculated for hot/cold days (using daily maximum temperature), heatwaves (also using daily maximum temperature), and hot/cold nights (using daily minimum temperature), as different non-optimal temperature hazards to investigate. If, however, daily maximum and minimum temperatures are not available, daily mean temperature can be used instead and outputs produced for hot days and nights together, as well as cold days and nights. The results tend to be similar between the various temperature variables, but with significant enough differences to warrant a separate exploration if possible.

The method has three main steps, as described below.

Stage 1: Fitting DLNMs

This first stage of the analysis fits a DLNM to the data for each locality to determine the temperature-mortality association, reported as a RR. Possible confounders are controlled for in the DLNM.

To be more precise, the DLNM fits a quasi-Poisson generalised linear model to the data, controlling for the possible confounders, the independent variables. The formula for the DLNM can be written as

$$death \sim cb + independent\ variables + ns(date, df)$$

where cb is a cross-basis matrix, and $ns(date, df)$ is a natural B-spline, the term which controls for seasonality (with 8 degrees of freedom) and long-term trends. This spline has degrees of freedom

$$df = (\text{degrees of freedom for seasonality} \times \text{number of years in the timeseries})$$

Long-term trends are controlled for to be able to assess the short-term impact of non-optimal temperature on health. Sensitivity analysis will indicate which independent variables to include in the final model, as described in section 3.4.

The cross-basis matrix is the parameterisation of a composite of two functions: a temperature-response and a lag-response function. The temperature-response function uses daily temperature in the locality as the exposure variable, and a B-spline function with knots at the 10th, 75th and 90th quantiles of daily temperature in that locality. The lag-response function uses a lag of 21 days (or 3 weeks for weekly aggregated data), and 3 equally spaced knots, as short-term health outcomes from

temperature can arise up to around 21 days after the exposure (Gasparrini, et al., 2015). The number of days in the lag can be adjusted if national literature indicates a need for adjustment.

The temperature-mortality association is subsequently estimated for each locality by calculating a predicted RR with the fitted DLNM where the reference value is taken to be the average of the timeseries' daily temperatures in that locality.

Stage 2: Calculating MMTs and OTRs

This second stage of the analysis calculates the MMT and OTR for each locality. The MMT is the reference point for calculating the AN and AR for non-optimal temperatures, those outside of the OTR.

MMT:

The calculation of local MMTs is done with a meta-analysis, pooling estimates of locality-specific temperature-mortality associations, and using these pooled estimates to re-calculate the local RR of temperature on mortality. This is to ensure localities with small daily death counts have enough information for precise estimates. The three steps for the calculation of the MMTs are:

- First, the estimated RRs for all localities are pooled using a multivariate random effects meta-analytical model with meta-predictors. Two meta-predictors are included to account for locality-specific effect modifiers: 1) locality-specific average of daily temperature in the timeseries, and 2) the locality-specific range size for daily temperature in the timeseries. The p-value of each meta-predictor's Wald statistic is also calculated to determine which meta-predictors are significant for their area.
- Second, derive the best linear unbiased prediction (BLUP) of the model coefficients for each locality from the coefficients of the meta-analytical model.
- Finally, derive the MMT and its percentile by predicting the RRs for temperature percentiles based on these BLUPs.

OTR:

The OTR for each locality is determined by the range of temperatures for which the newly predicted RR of death is between 1 and 1.1.

If the RR goes below 1 in the tails, the spatial scale can be increased. For example, it can be changed from the local to the provincial scale.

Stage 3: Calculating ANs and ARs

The third stage of the analysis calculates the AN and AR of mortality from non-optimal temperatures, and their respective confidence intervals.

The AN and AR for each locality are calculated for moderate-intensity extreme temperatures, and for high-intensity extreme temperatures to identify differences in outcomes as temperatures go further into the extremes. Only those days t_i whose temperature x_{t_i} is in the relevant range are included as contributable components in the calculations for each temperature range.

A locality's AN for each day in a year is given by:

$$AN_{x,t} = \left(1 - \exp \left(- \sum_{l=l_0}^L \beta_{x_t,l} \right) \right) \times \sum_{l=l_0}^L \frac{n_{t+l}}{L - l_0 + 1}$$

where $\beta_{x_t,l}$ are the contributions to the risk of death from temperature x_t on day t , and n_t is the number of deaths on day t . Since the impact of the temperature x_t on day t is lagged, the contributions to the risk are summed over the lag period, that is, over days $t + l_0, \dots, t + L$ where l_0 and L are the minimum and maximum lags respectively.

The locality's AN for the year is calculated by adding the AN contribution from each day in the year:

$$AN_{tot} = \sum_{i=1}^m AN_{x,t_i}$$

where $i = t_1, \dots, t_m$ are all the days in the year. The final AN, given for the locality as the result for the output year, is taken to be the average of the output year's AN and of the yearly AN of the four years before that. This averaging over 5 years is done to smooth out the estimates and be able to identify trends in time more easily.

Such averaging is also done for the output year's AR. A locality's AR of deaths per 100,000 population for the year is calculated from the 5-year averaged AN as:

$$AR_{tot,y_{output}} = \frac{\sum_{i=0}^5 AN_{tot,y_{output}-i}}{\sum_{i=0}^5 P_{y_{output}-i}/5} \times 100,000$$

where P_y is the estimated size of the locality's population in year y .

The confidence intervals for both the AN and AR are given by Monte Carlo simulations with 1000 simulations.

3.4. Sensitivity analysis

Test the following modelling choices through sensitivity analysis:

Modelling choice	Description	How
Pollutant to use for air pollution concentration levels	Choose between the use of - PM10, - PM2.5, - NO2, and - O3 as the air pollution variable. These pollutants have been identified as having a joint effect with heat on mortality (Stafoggia, et al., 2023).	If there is more than one air pollution variable with a high proportion of data completeness, choose the pollutant which has highest correlation with mortality.
Independent variables in the DLNM formula	Test whether each of the following possible independent variables should be included in the DLNM formula in stage 1, if data is available: - Humidity - Air pollution	The AIC of the meta-analytical model fitted in stage 2 can be compared between combinations of independent variables included in the DLNM. The combination of independent variables with the lowest AIC is chosen as the final input for the DLNM.

	• Wind speed • Air pressure • Day of the week	
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3.5. Communication & presentation

The results from this indicator are best presented by a combination of statistical tables and graphs. Table 2 describes how each output from this indicator can be presented, with examples of graphs drawn from the ONS' publication on [Climate-related mortality in England and Wales](#).

Table 2: presentation and interpretation of analytical outputs

Output	Presentation
The RR of death by temperature	Present the RRs in statistical tables. Their respective estimates should be included for each locality and for each disaggregation for the whole country. These tables should be repeated for each of the temperature hazards. Line graphs can be used to represent differences in RR patterns between localities and disaggregation groups: • A line graph of the RR of the various temperature hazards by locality. • A line graph of the RR of death from the various temperature hazards by disaggregation group. Headline results could be quoted as: • “Both very low and very high temperatures had higher mortality risk, with temperatures below xx and above xx degrees Celsius representing the greatest risk, in <i>reference area</i> in years yy to yy.” • “The highest mortality risk was in <i>reference area</i> for temperatures exceeding/below xx degrees Celsius, where mortality risk was xx times the risk at optimal temperatures.” The following is an example of graphs to represent the RR output:

The AN and AR of death from non-optimal temperature

Present the ARs and ANs in statistical tables. Their respective estimates and confidence intervals should be included for each locality and for each disaggregation for the whole country. These tables should be repeated for each of the temperature hazards.

Timeseries plots and choropleth maps can be used to represent changes over time:

- A line or dot plot of the AR from the various temperature hazards over the years by region or for the country.
- A choropleth map of the AR in each locality over the years.

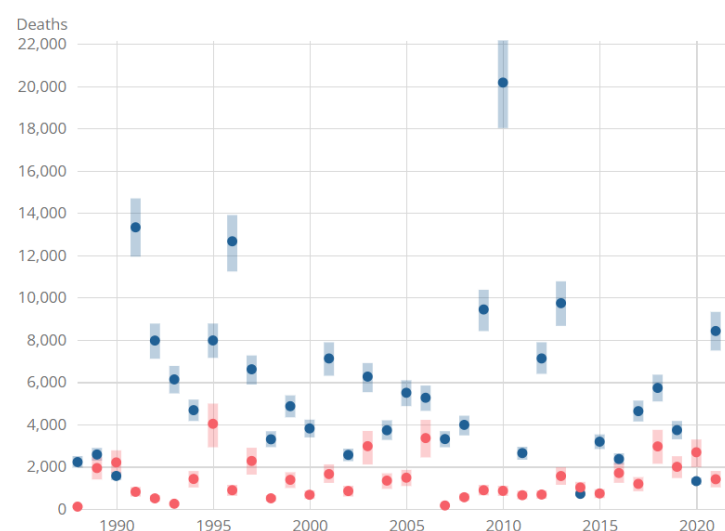
Headline results could be quoted as:

- “An estimated xx deaths (95% confidence interval: xx to xx) in *reference area* were associated with *temperature hazard* from yy to yy.”
- “There was some indication that deaths from *temperature hazard* have increased over recent years; in *output year* an estimated xx deaths (95% confidence interval: xx to xx) were associated with *temperature hazard* in *reference area*.”

A dot plot of the AR of the hot and cold days would be similar to this example dot plot of ANs in England:

Estimated temperature related deaths, England, 1988 to 2022

● Coldest days (with 95% confidence interval)
● Hottest days (with 95% confidence interval)



3.6. Limitations & interpretation

The first main outcome for this indicator is the RR of death by temperature. It indicates how the risk of mortality at a particular temperature compares with the risk at the MMT, i.e. the likelihood of an individual dying during, or shortly after, exposure to a certain temperature compared to the likelihood of death at the MMT. When a temperature has a RR of 1, this means there is neither an increase nor a decrease in the likelihood of the individual dying during, or shortly after, exposure to that temperature. A RR of 1.1 denotes a 10% increase in mortality risk compared to the MMT. The

RR can also help identify turning points at which temperature quickly becomes more dangerous for that locality.

The AN and the AR are the estimated number of deaths and rate per 100,000 population, respectively, that would not have occurred if there had been no exposure to non-optimal temperatures within the determined time lag before death. These are also the AN and AR of deaths which cannot be explained by the independent variables that have been accounted for in the model. However, if a true confounder is not included in the model, the deaths attributed to non-optimal temperatures can be biased.

The AN and AR are based on the temperatures experienced in a particular year and any change in climate towards more extreme temperatures would likely lead to an increase in attributable deaths. For example, if there are more days with temperatures exceeding thresholds where the mortality risk is high, the AN and AR will rise. Therefore, a decrease in AN or AR does not necessarily indicate better adaptation to non-optimal temperatures from one year to the next. It may simply be due to there having been less exposure to non-optimal temperatures that year. It is, therefore, important to interpret this indicator alongside the topic area's exposure indicators.

When making comparisons between localities and disaggregated groups, the ARs should be used since it considers the differences in population sizes.

Comparing the AR of disaggregated groups can help identify possible social confounding factors. However, some disaggregation variables may be correlated with one another, e.g. ethnicity, socio-economic group and occupation. Strong correlations should be reported in the output to inform possible reasons for differences and therefore, most urgent adaptation needs. This may relate to their respective lifestyles, health statuses, and exposure to non-optimal temperatures.

A limitation to note with regards to the disaggregations is that data is not available to disaggregate by each major effect modifier. For example, those with underlying health conditions are most probably more vulnerable to the effects of extreme temperatures. Most countries do not, however, have data on this and including it as a disaggregation would not be realistic. Additionally, there is still limited research on the social factors affecting temperature vulnerability, therefore, not all effect modifiers are known.

To account for limitations in data completeness and accuracy, and the ensuing uncertainty in the indicator estimates, it is important to report their uncertainty (the confidence intervals) and to consider them in the interpretation of variations in the output. Among many possible reasons, such variations may be related to differences in data collection or data sources.

A limitation with the method is that outdoor temperature does not capture a population's true exposure to non-optimal temperatures. For example, personal levels of exposure to heat in the home, which may vary depending on the presence of cooling systems, are not captured through ambient temperature data. This is why the interpretation of this indicator should be complemented with information from the framework topic area's vulnerability, exposure, and adaptation indicators.

A limitation with the use of weekly data is that weekly data models tend to underestimate attributable deaths from all temperatures by roughly 10% compared to daily data models – about 8.5% for cold attributable mortality and 21.5% for heat attributable mortality (Ballester, et al., 2023). However, the underestimation of mortality attributed to extreme temperature events is only around 5%. When comparing attributable mortality across years, this underestimation is not an issue because the absolute differences between weekly and daily data model estimates tend to remain relatively constant over the years. If, however, outcomes are compared between a daily and a weekly data model, this underestimation should be considered in the interpretation.

4. Further reading

[Mortality risk attributable to high and low ambient temperature: a multicountry observational study - The Lancet](#)

[Attributable risk from distributed lag models | BMC Medical Research Methodology | Full Text \(biomedcentral.com\)](#)

[Time series regression studies in environmental epidemiology - PMC \(nih.gov\)](#)

[Reducing and meta-analysing estimates from distributed lag non-linear models | BMC Medical Research Methodology | Full Text \(biomedcentral.com\)](#)

[Modeling exposure–lag–response associations with distributed lag non-linear models - PMC \(nih.gov\)](#)

[Models for the Relationship Between Ambient Temperature and Daily Mortality : Epidemiology \(lww.com\)](#)

[An INDEPTH look at global data collection – The Lancet](#)

[ERA5 global atmospheric reanalysis](#)

[Air Quality Modelling Products – 4 Earth Intelligence](#)

5. References

Ballester, J. et al., 2023. The effect of temporal data aggregation to assess the impact of changing temperatures in Europe: an epidemiological modelling study. *The Lancet Regional Health - Europe*, Issue 100779.

Gasparrini, A. et al., 2015. Mortality risk attributable to high and low ambient temperature: a multicountry observational study. *Lancet*, Volume 386, p. 369–75.

Stafoggia, M. et al., 2023. Joint effect of heat and air pollution on mortality in 620 cities of 36 countries. *Environment International*, Volume 181.